



CN's approach to ATGMS operations and development



DALLAS, TX



Program Introduction

CN's ATIP Program began in 2019 with instrumentation of eight boxcars.

ATIP boxcars are operated within Revenue Service

- Scheduled inspections
- Fast interchanges in/out of yards
- No operating overhead
- No impact on network capacity

ATIP boxcars provide dynamic geometry measurements

- Each car weighs between 220,000-230,000 pounds
- Better insights into health of network



10

ATIP cars
currently in
CN fleet

Built with modular capacity for new technology expansion

- Rail wear measuring systems
- Vision based inspection systems (Ties, OTM, Ballast, Joints, etc.)
- Ground Penetrating Radar inspection systems

Breadth of CN ATIP Program



Network Layout, Inspection Velocity Advantage

CN’s network design, combined with transportation support, have resulted in industry-leading velocity and network coverage with its ATIP Fleet.

Technology Growth & Development

CN has deployed some of the industry’s most extensive inspection technologies across its ATIP Fleet and continues to explore additional measurement capabilities.

Miles of Track Inspected per Year					
	2015	2016	2017	2018	2019
Manned HGIV (Core Lines)	51,000	54,000	35,000	43,000	41,000
Manned HGIV (Non-Core Lines)	30,000	29,000	19,000	24,000	22,000
Grand Total	81,000	83,000	54,000	67,000	63,000

Miles of Track Inspected per Year						
	2020	2021	2022	2023	2024	2025
Manned HGIV (Core Lines)	41,000	32,000	35,000	26,000	22,000	24,000
Manned HGIV (Non-Core Lines)	24,000	21,000	19,000	18,000	18,000	16,000
ATIP (Core Lines)	503,000	797,000	842,000	855,000	829,000	711,000
ATIP (Non-Core Lines)	6,000	25,000	47,000	48,000	49,000	68,000
Grand Total	533,000	875,000	943,000	947,000	918,000	819,000



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CN ATIP Data Utilization



Reduction in Defect Latency

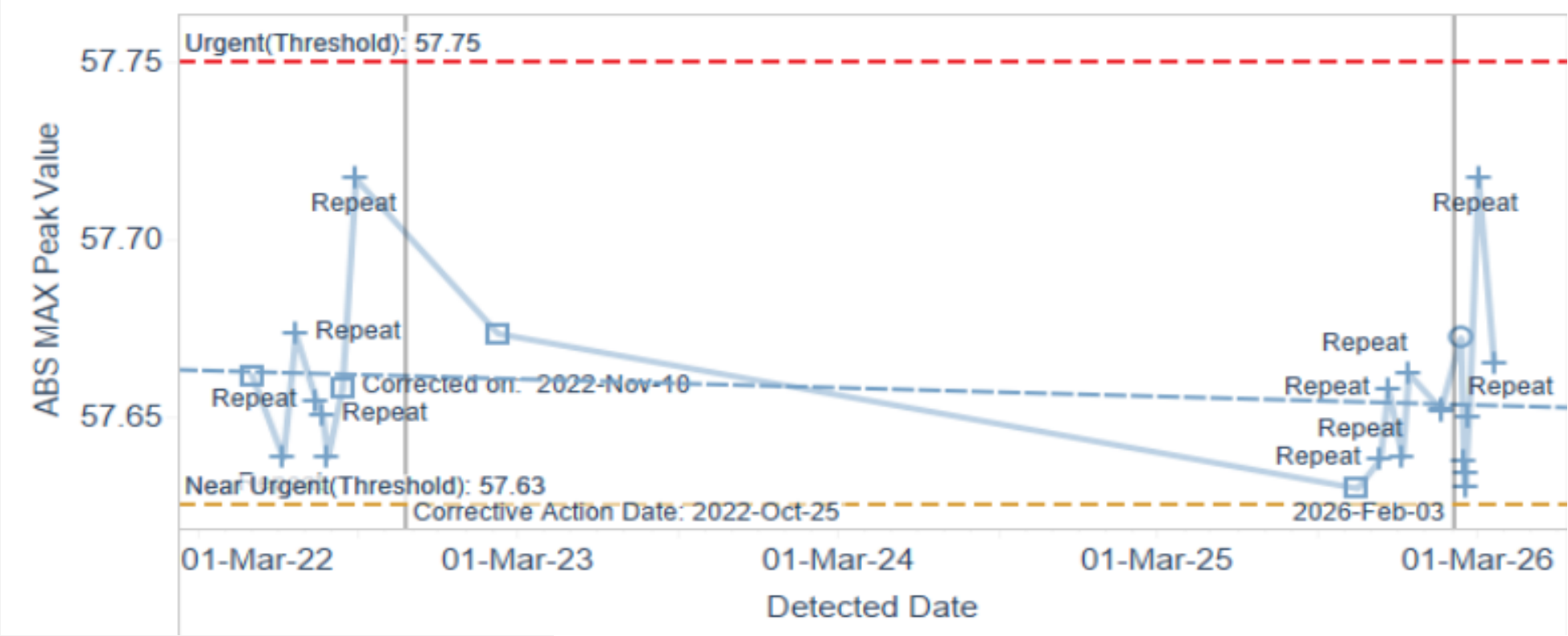
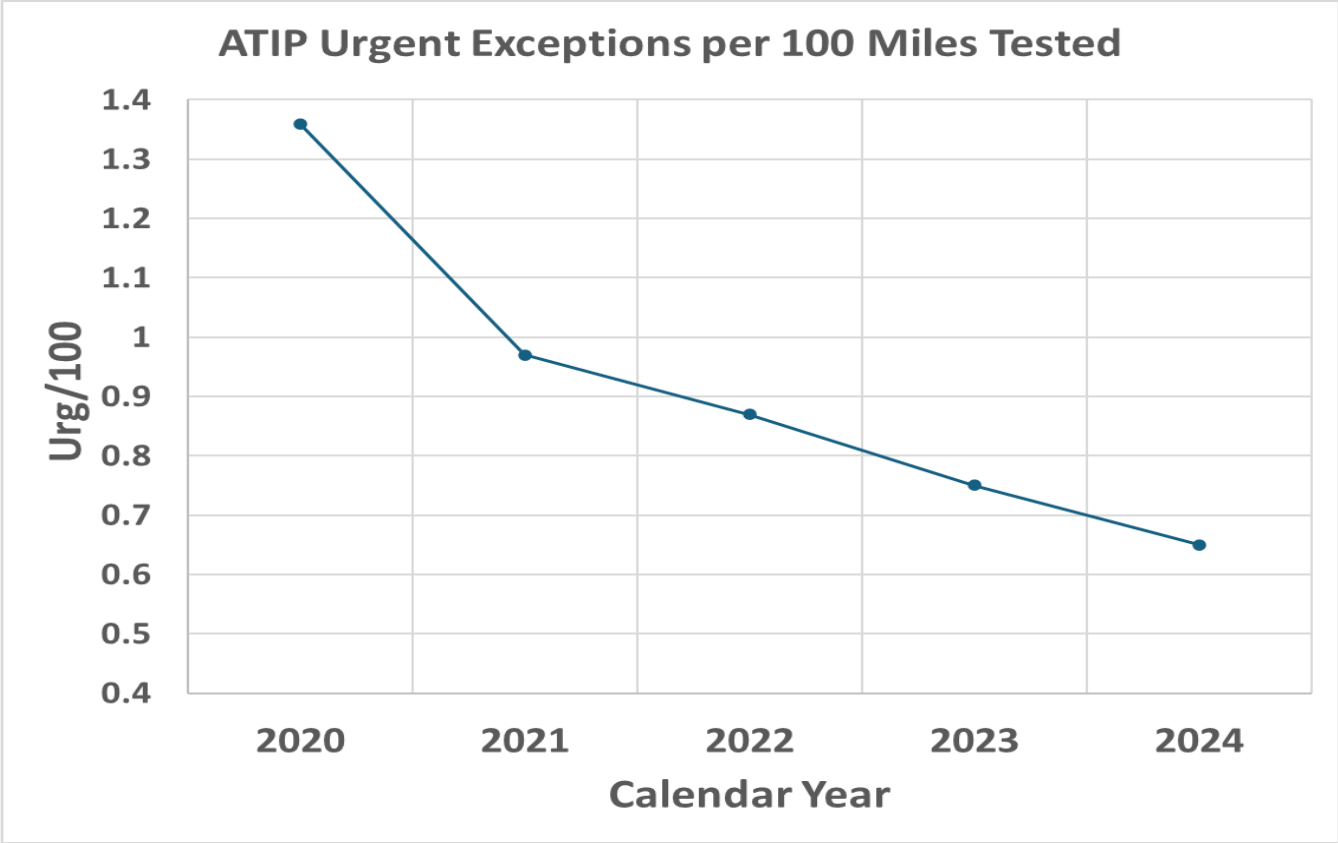
- Increased HGIV testing frequency and velocity enables earlier detection of track conditions
- Confirms repairs are effective, reducing recurrence of defects

Condition Growth Monitoring

- High inspection volume enables tracking of defect progression over time
- Supports optimized and prioritized maintenance planning

Performance Metrics Development

- Key metrics enable route and corridor level evaluation of track health
- Drives data informed decision making across the network



CN ATIP Data Utilization



Capital Program Building

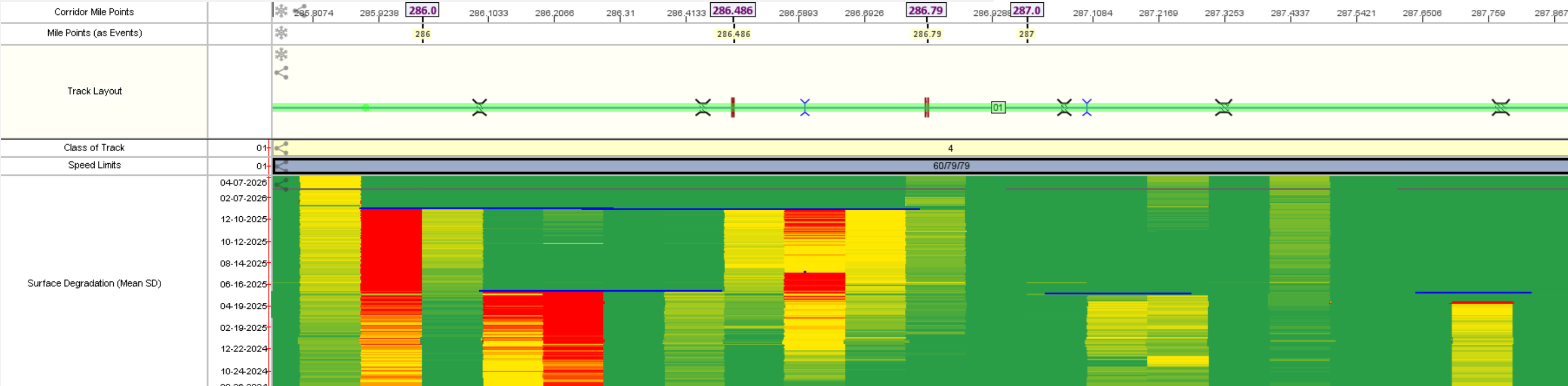
- Corridor and network level aggregation to visualize and prioritize large-scale repairs

Targeted Surfacing Repairs

- Identify growth trends to coordinate interventions
- Evaluate repair effectiveness to ensure sustained performance

Surface Quality Scores per Engineering Hierarchy

vision Name	Track ID	LA Start Location	LA End Location	Length (mi)	Surface TQI 62 Mean Level Latest	Surface TQI Band Latest	AVG Current Year Surface TQI 62 Mean	AVG Current Year Surface TQI Band	AVG Prior Year Surface TQI 62 Mean	AVG Prior Year Surface TQI Band	
0	01	104.00	104.10	0.10	0.44	GOOD	0.04	GOOD	0.70	GOOD	
0	01	104.10	104.21	0.10	0.46	GOOD	0.59	GOOD	0.57	GOOD	
0	01	104.21	104.31	0.10	0.46	GOOD	0.94	GOOD	0.70	GOOD	
0	01	104.31	104.41	0.10	0.61	GOOD	2.69	POOR	2.14	POOR	
0	01	104.41	104.52	0.10	1.93	FAIR	3.23	BAD	2.82	POOR	
0	01	104.52	104.62	0.10	3.60	BAD	3.98	BAD	3.85	BAD	
0	01	104.62	104.72	0.10	0.54	GOOD	1.19	FAIR	1.07	FAIR	
0	01	104.72	104.82	0.10	0.76	GOOD	3.70	BAD	3.93	BAD	
0	01	104.82	104.91	0.10	0.78	GOOD	3.65	BAD	3.56	BAD	
0	01	104.91	105.00	0.10	0.61	GOOD	2.19	POOR	1.82	FAIR	
0	01	105.00	105.10	0.10	0.65	GOOD	0.81	GOOD	0.87	GOOD	





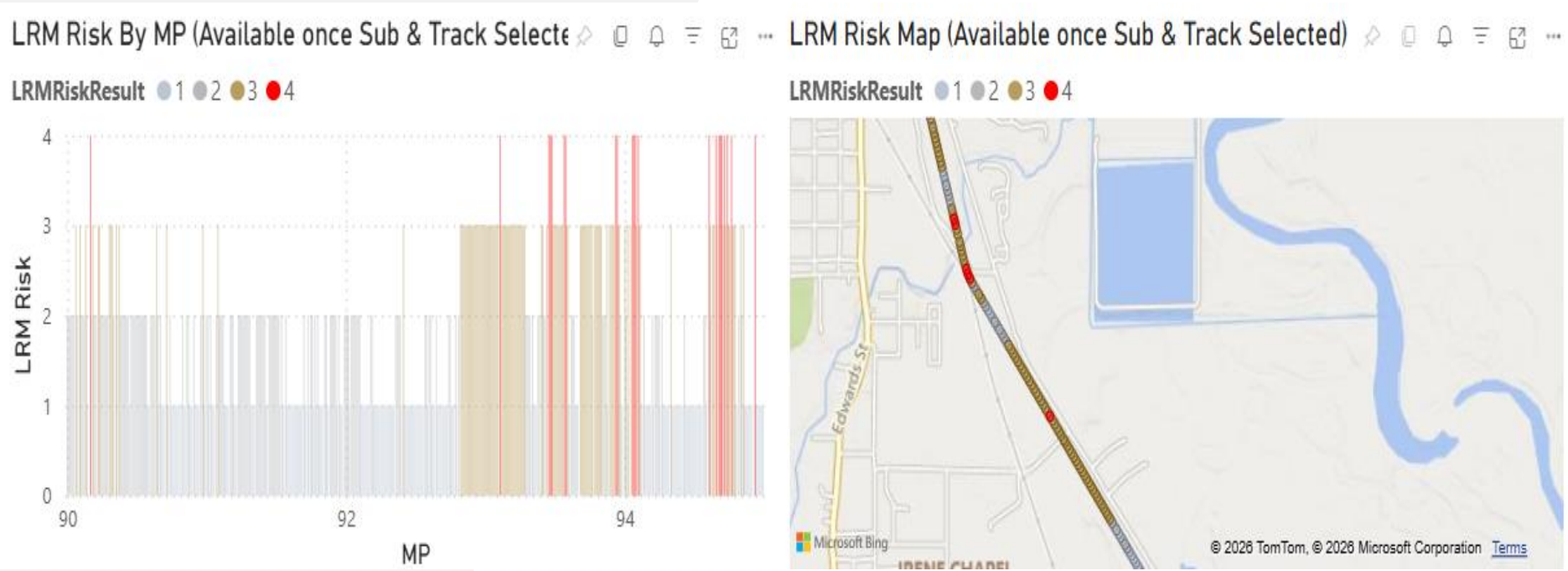
Advanced Development



Solving Complex Operational Challenges

- Leveraging ATIP’s extensive measurement data to develop solutions for industry wide challenges
- Longitudinal Rail Movement Model
 - Analyze track components at the network level to identify high-risk locations
 - Enable targeted inspections and scalable maintenance strategies
- Ice Jacking Reporting
 - Identify and highlight locations with track conditions susceptible to ice jacking

Track	MP	LRM Risk	Track Structure Risk	TrainGradeGeoResult	Anchor Movement Flag	Box Anchor Flag	Crib Ballast Flag	Tie End Ballast Flag
01	5.24	2	2	2	3	3	1	1
01	5.25	3	3	2	2	5	1	3
01	5.26	3	3	2	3	4	1	3
01	5.27	3	3	2	2	5	1	3
01	5.28	2	2	2	4	3	1	2
01	5.29	3	3	2	3	4	1	1
01	5.30	2	2	2	2	3	1	1
01	5.31	3	3	2	3	4	1	1
01	5.32	3	3	2	2	4	1	3
01	5.33	3	3	2	3	5	1	3
01	5.34	2	2	2	2	3	1	3
01	5.35	2	2	2	3	3	1	3
01	5.36	2	2	2	2	2	1	3
01	5.37	1	1	2	3	2	1	1
01	5.38	1	1	2	2	2	1	1
01	5.39	1	1	2	2	2	1	1
01	5.40	1	1	2	2	1	1	1
01	5.41	1	1	2	2	2	1	1





Benefits with CN Approach



Testing Velocity/Frequency

- Increased testing frequency improves visibility, trend analysis and maintenance prioritization
- Delivered with no operational impact to the network
- Enables earlier detection of track conditions and more effective repairs

Extensive Technology Suite

- ATIP fleet's advanced inspection capabilities provide frontline engineering with unmatched visibility into track health
- Ongoing expansion and innovation to address evolving industry challenges

**Increased, efficient inspections.
Diverse measurement capabilities.**



Lessons Learned from ATIP Program

Strong Location Database (GIS)

- Accurate geospatial referencing enables precision, repeatability, and reliability of inspection systems
- Ongoing expansion of CN's GIS database enhances ATIP program capabilities

Scalability and Modularity

- Modular ATIP architecture enables rapid integration of new technologies
- Supports future advancement with minimal disruption to operations



Acknowledgements

ENSCO

Vehicle layout design, fleet build support, measurement systems

Tetra Tech

Vehicle layout design, power system design and supply, fleet build support, measurement systems on some of the fleet

The Escanaba and Lake Superior Railroad

Vehicle buildout, including repair/rebuild of selected boxcars, custom modifications, installation facility support

Bentley Systems

Asset Management Solution